

High-Power Superconducting Electron Linacs for Commercial Applications

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High-Power SRF Linacs for Commercial Applications

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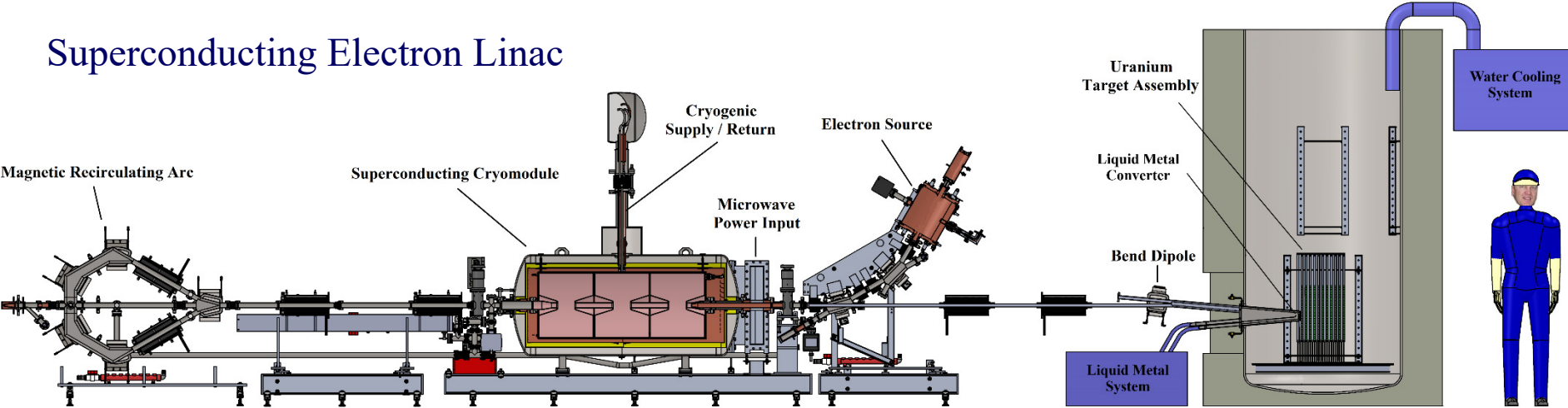
- Radioisotope production with high-power electron beams
 - Isotope production targets
 - Radiochemical separation techniques
- Commercial superconducting accelerator technology
 - Continuous-wave electron guns
 - Liquid helium refrigerators
 - Microwave sources
 - SRF cavities and cryomodules
 - Recirculating beamline designs



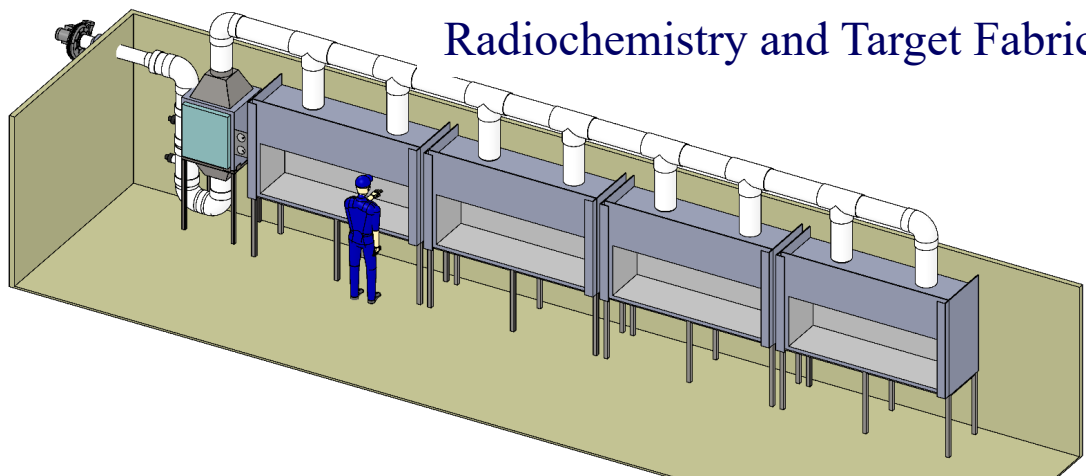
Radioisotope Production with High-Power Electron Beams

Isotope Production Target Station

Superconducting Electron Linac



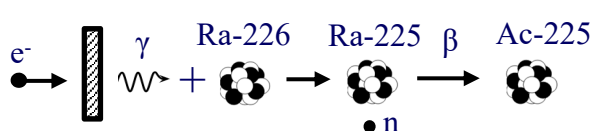
Radiochemistry and Target Fabrication





Niowave manufactures radioisotopes from radium and uranium

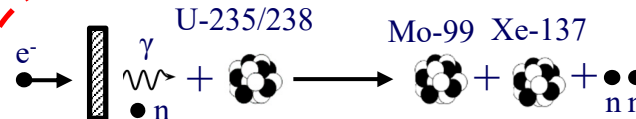
- Uses a superconducting electron linear accelerator without the need for a nuclear reactor



Radium targets
(Ra-226)

Ac-225	10 d
Rn-222	3.8 d
Po-210	138 d
Bi-210	5.0 d
Bi-213	46 m
Bi-214	20 m
Pb-214	27 m

α emitters
100 mCi/wk
Available in 2021



Uranium targets
(U-235, U-238)

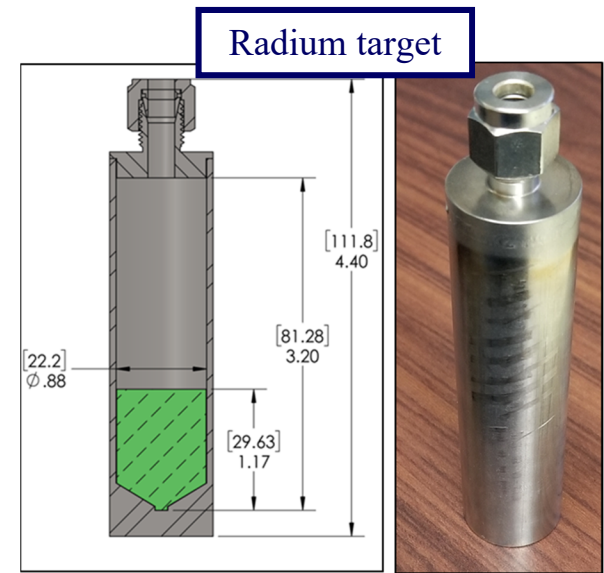
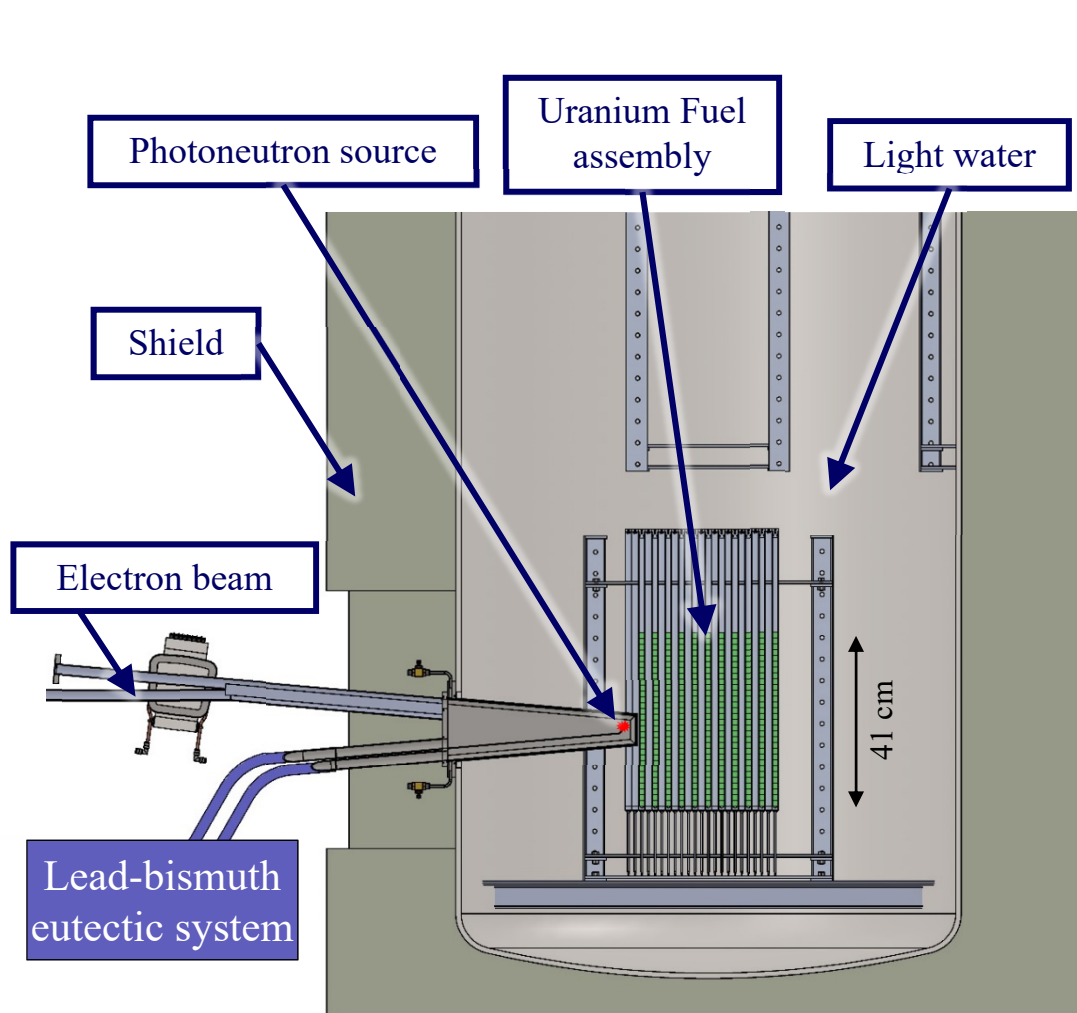
Mo-99	Pr-143
Sr-89	Pr-144
Y-90	Nd-147
Te-127	Pm-149
I-131	many more
Xe-133	

β emitters
10 Ci/wk
Available in 2021

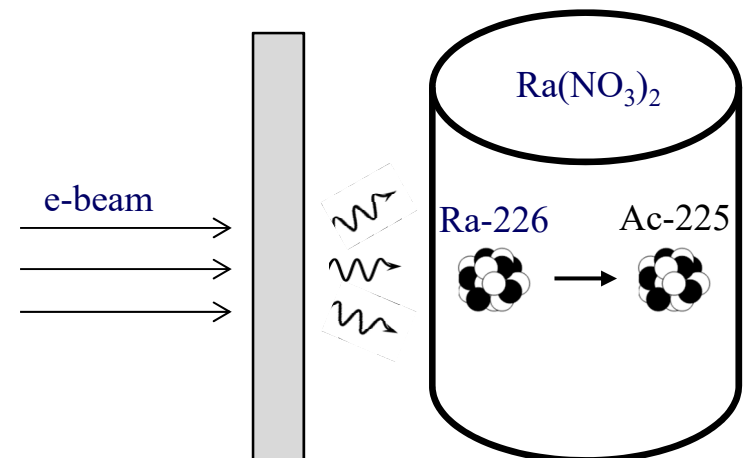


Isotope Production Targets

U_3O_8 fuel rods interchangeable with $Ra(NO_3)_2$ solution

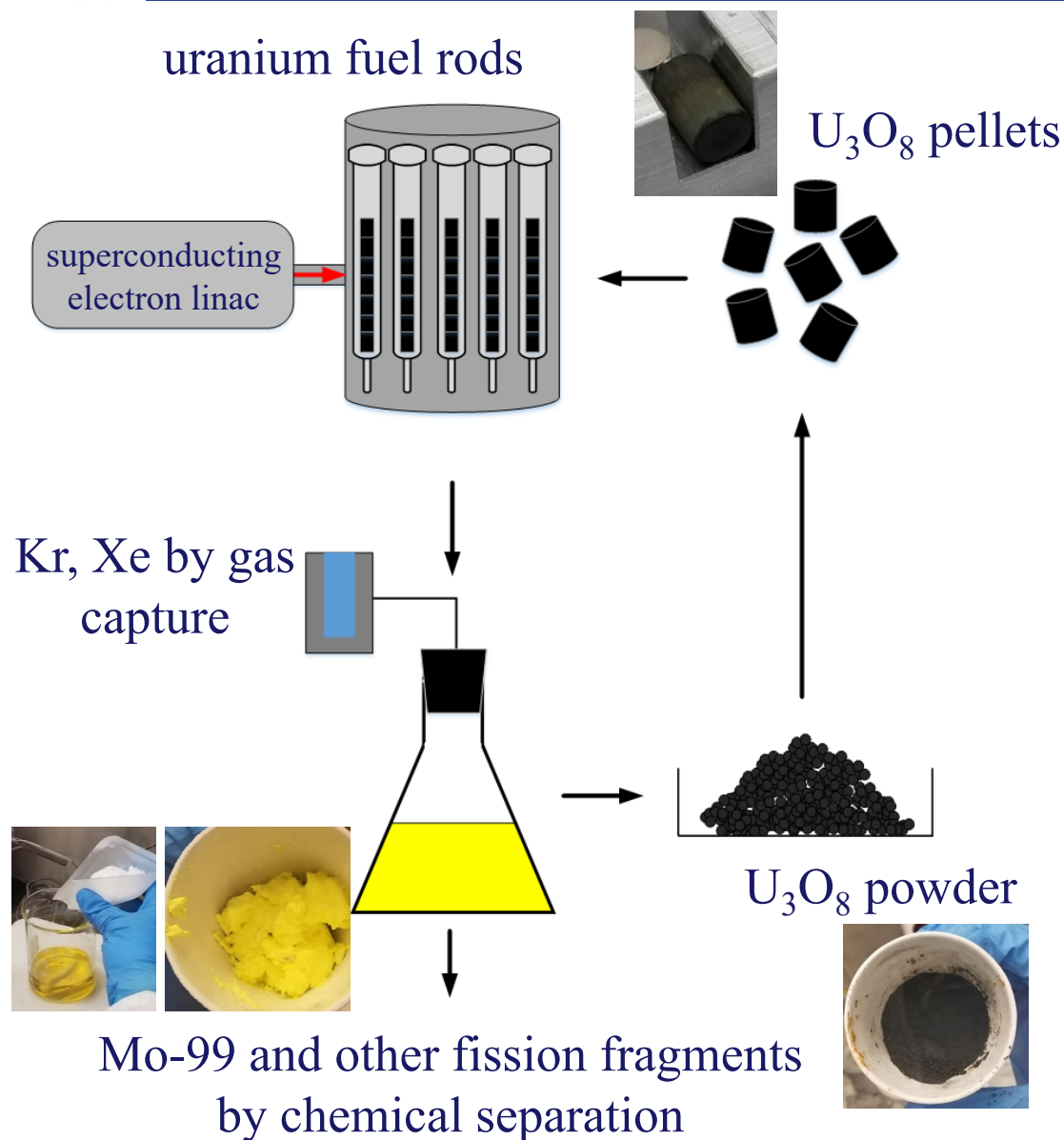


Converter

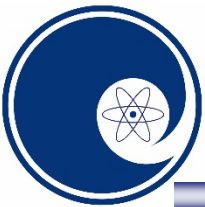




Radiochemical Separations

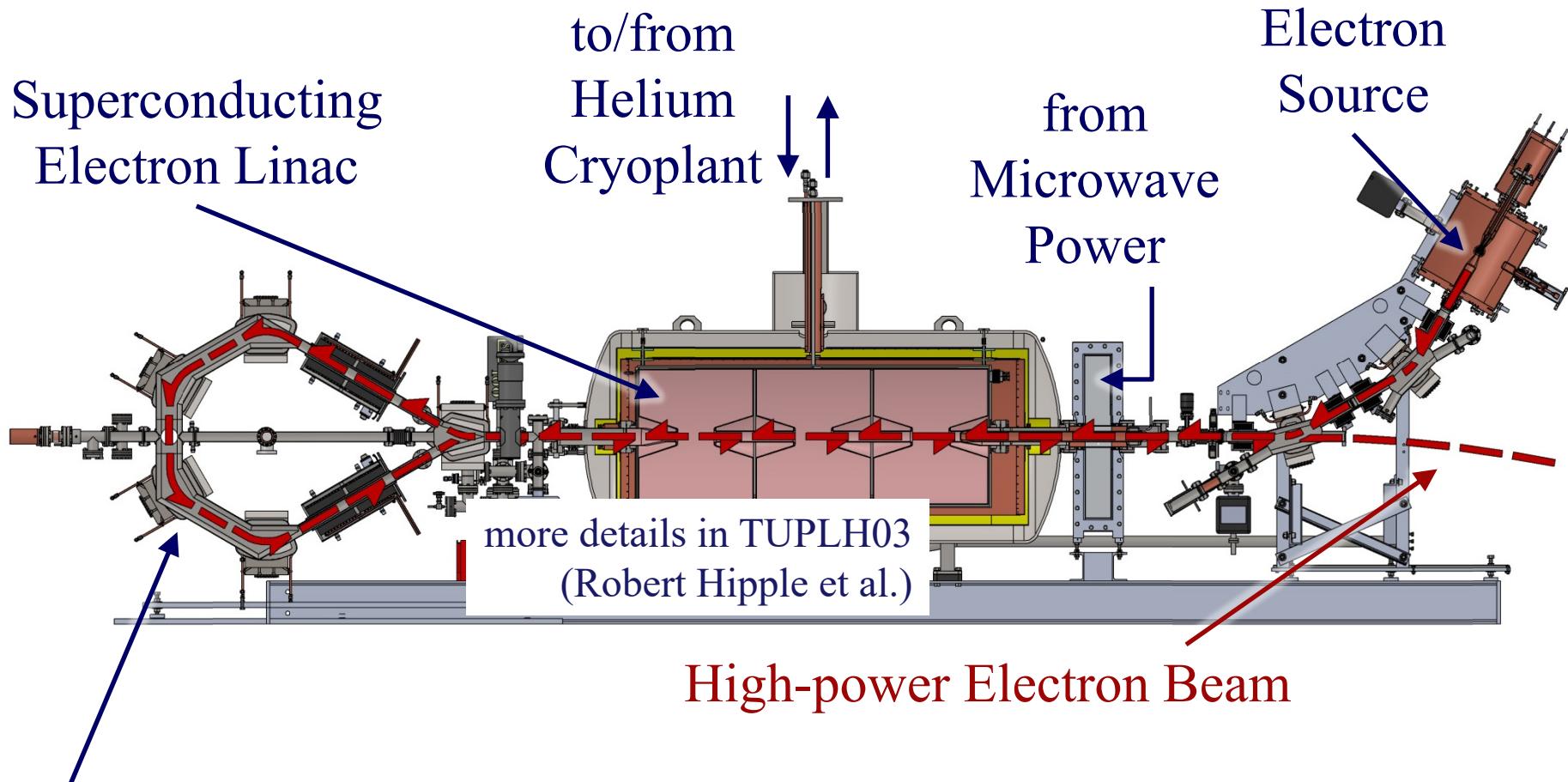


Separation facilities include a closed-loop uranium fuel cycle. Radioisotopes are separated and the uranium is converted back into fuel pellets. (See WEPLM20 by Bill Peters et al. for more details)

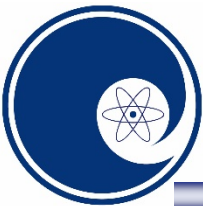


Two-pass Accelerator Design

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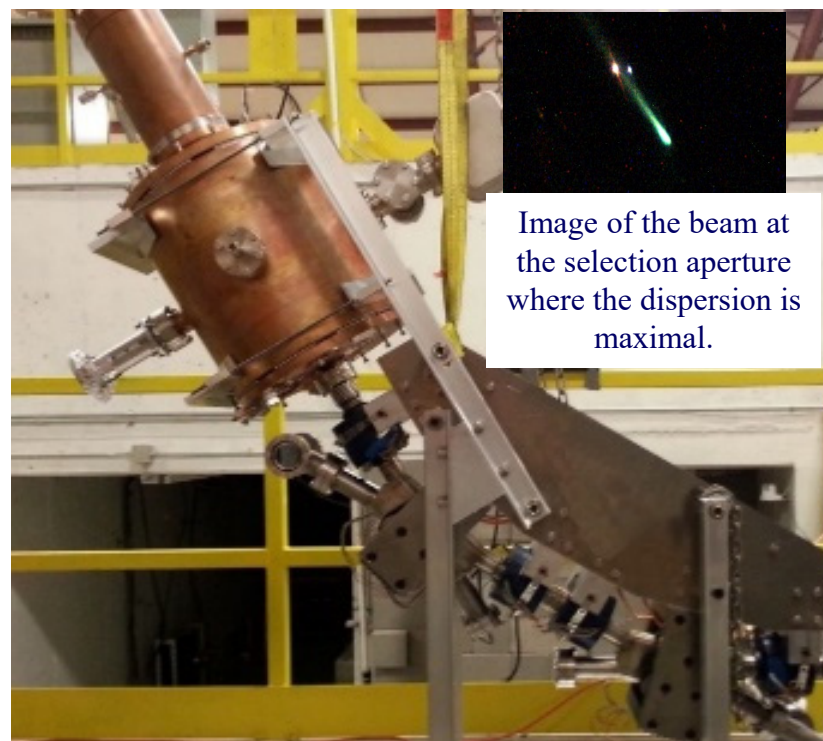
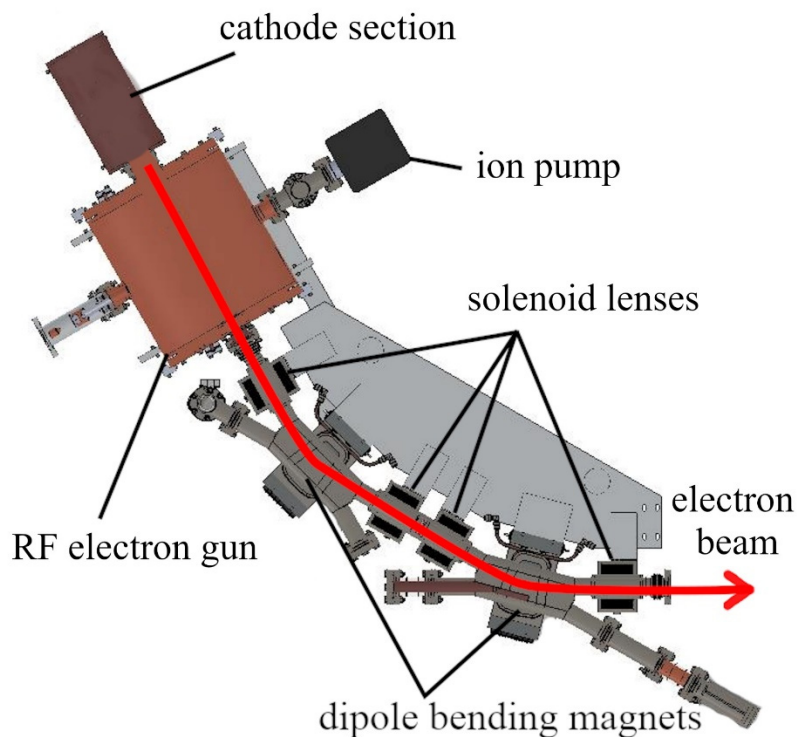
In this design, a magnetic arc (at left) brings the beam through the accelerator a second time, reducing costs for the cryomodule and refrigerator.



Thermionic Gun and DBA [1]

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- thermionic cathode takes advantage of high-duty cycle SRF operation (10%-100%)
- electron bunches from the cathode are shortened by gating fields (DC, fundamental RF, harmonic RF) at the cathode

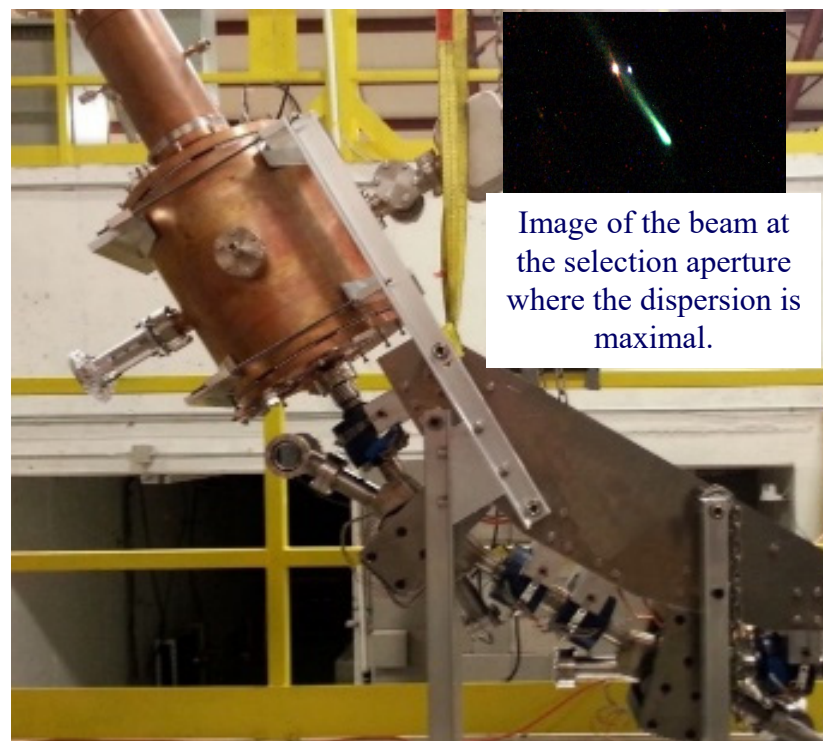
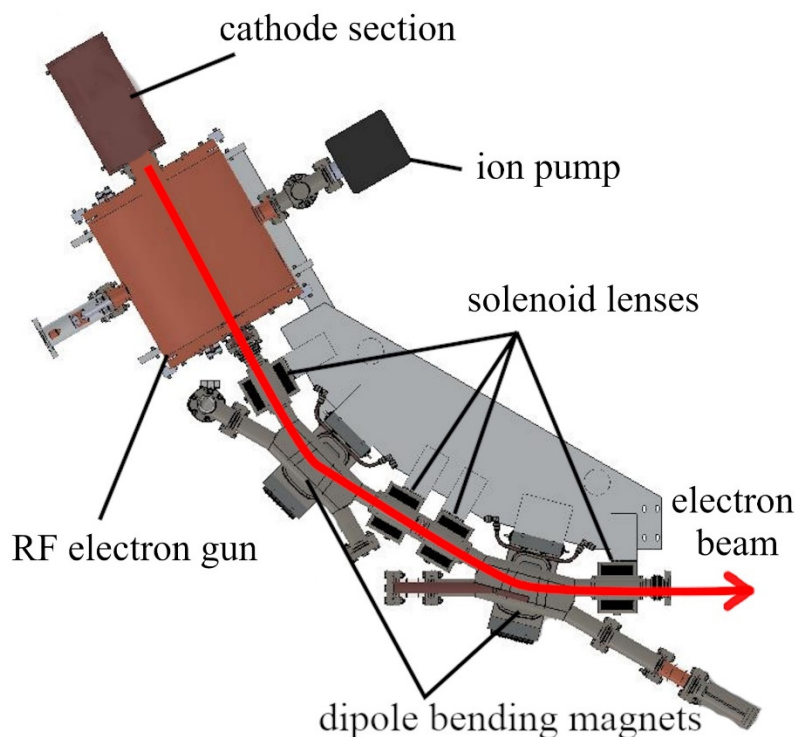


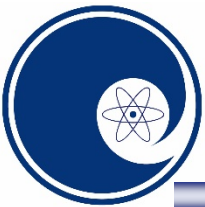


Thermionic Gun and DBA [2]

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- bunches are further filtered in a solenoid-based double-bend achromat
- more details in TUPLH03 (Robert Hipple et al.) on the interesting issues involved in using solenoids to do this



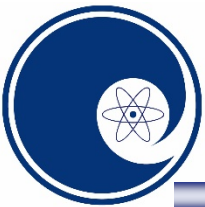


Helium Refrigerator/Cryocooler Systems

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- 51 W and 114 W helium refrigerators in operation
- Developed with Linde – Collins cycle, with robust reciprocating piston expander
- Vacuum insulated, nitrogen shielded, low-loss cryolines
- Modular cryolines allow quick switch between linac tests
- Long term operation ready:
 - 5000 hr. major maintenance interval
 - No warmup for short term maintenance

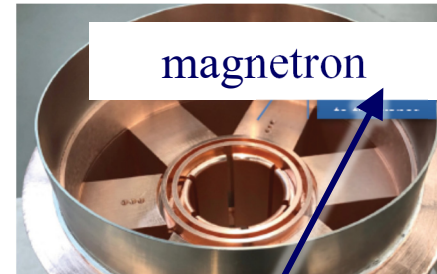




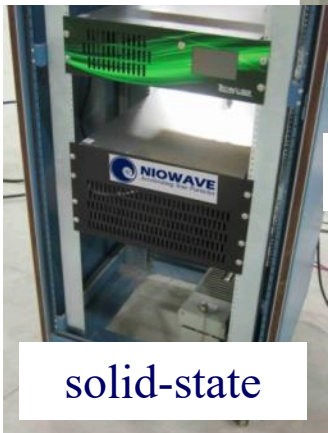
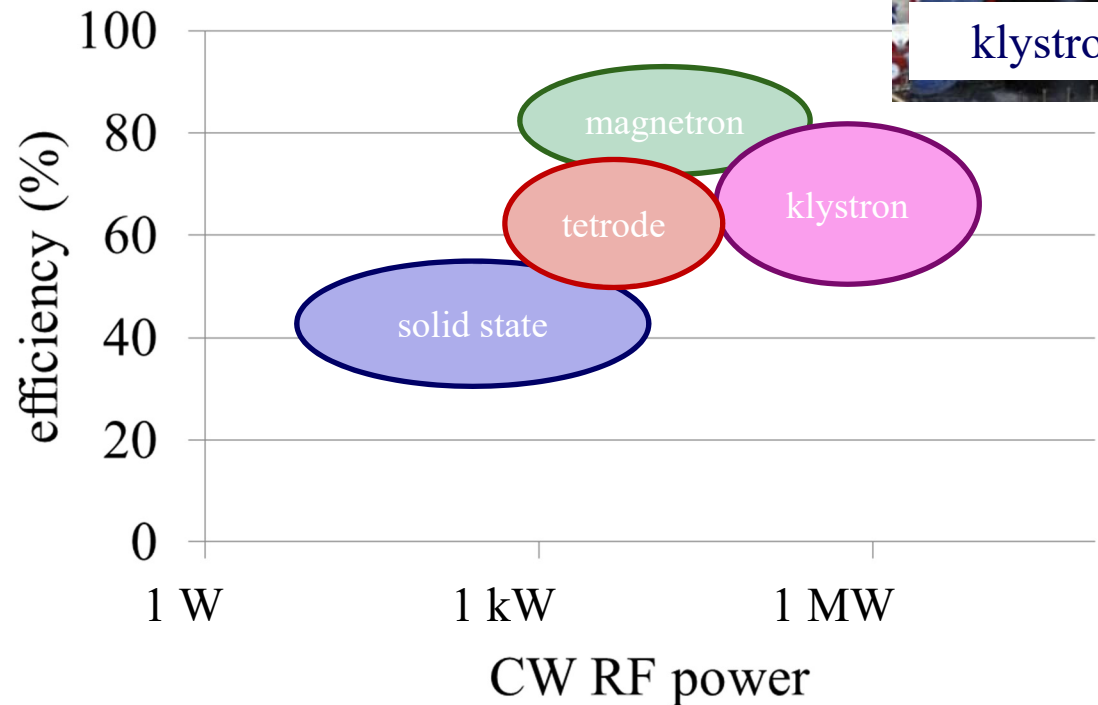
Microwave Power

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- Solid-state supplies to 5 kW
- Tetrode amplifier to 60 kW
- Magnetrons >100 kW
- Klystrons to >1 MW



see WEPLM58 (Mike Neubauer et al., Muons, Inc.)





SRF Cavities and Cryomodules

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Superconducting cavities
in specialized geometries



Cryomodules

Niowave started out building cavities and cryomodules for this community, both through direct contracts and collaborative SBIR/STTR projects.



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Cryomodules



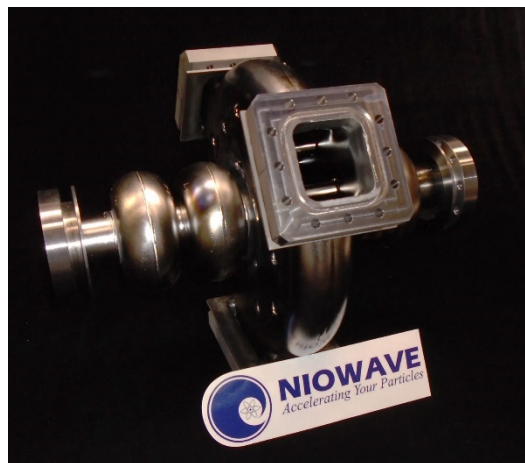
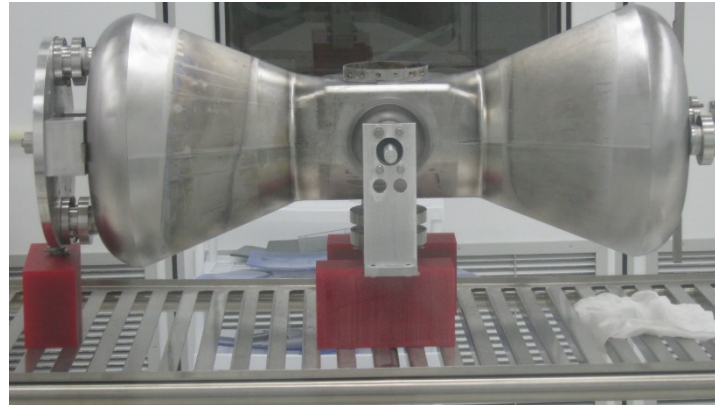
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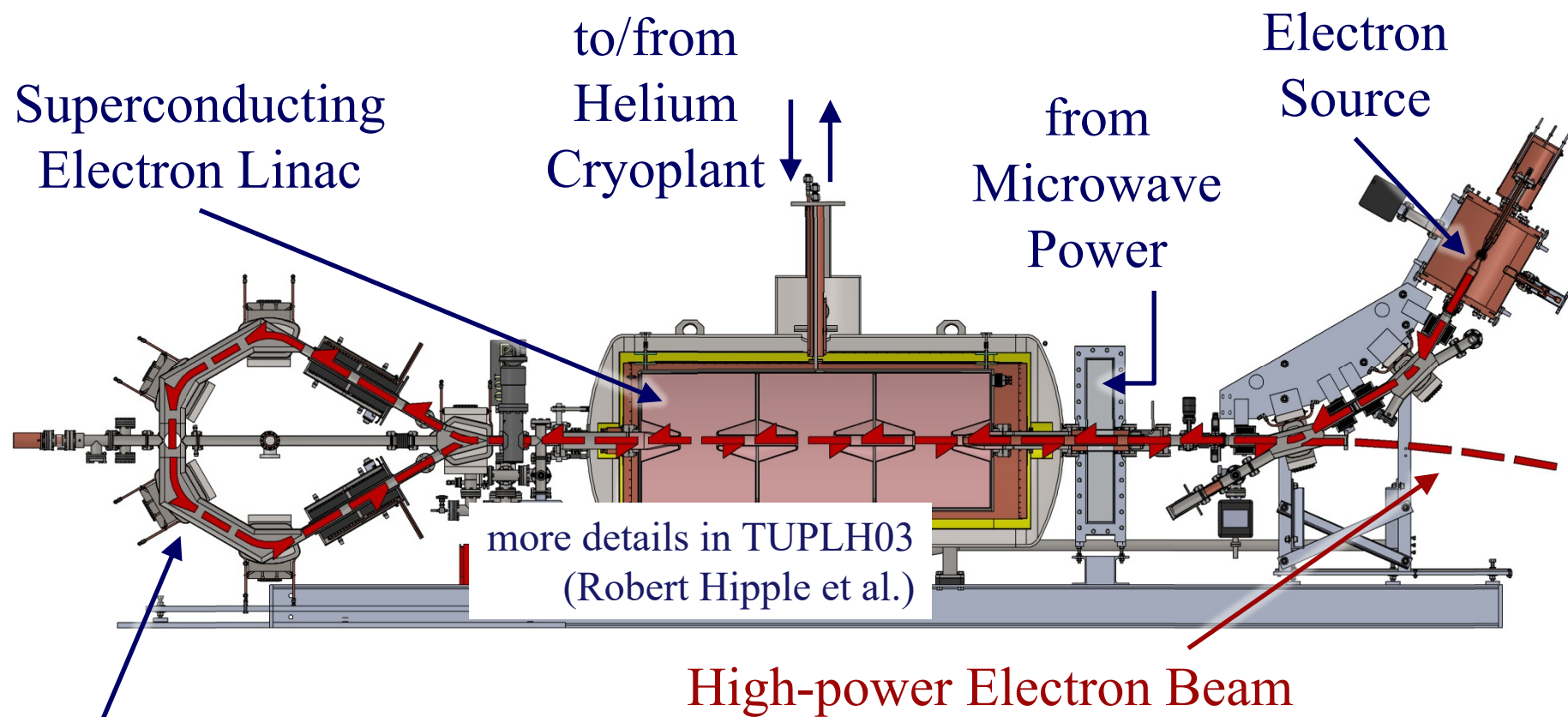


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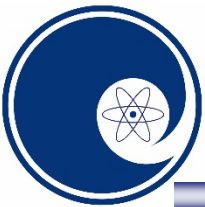


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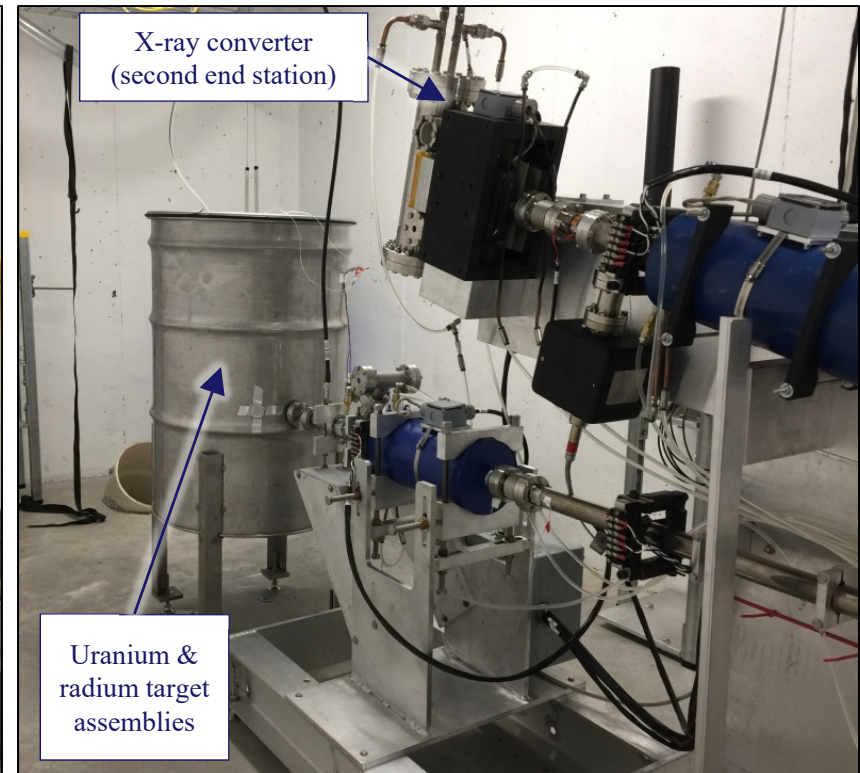
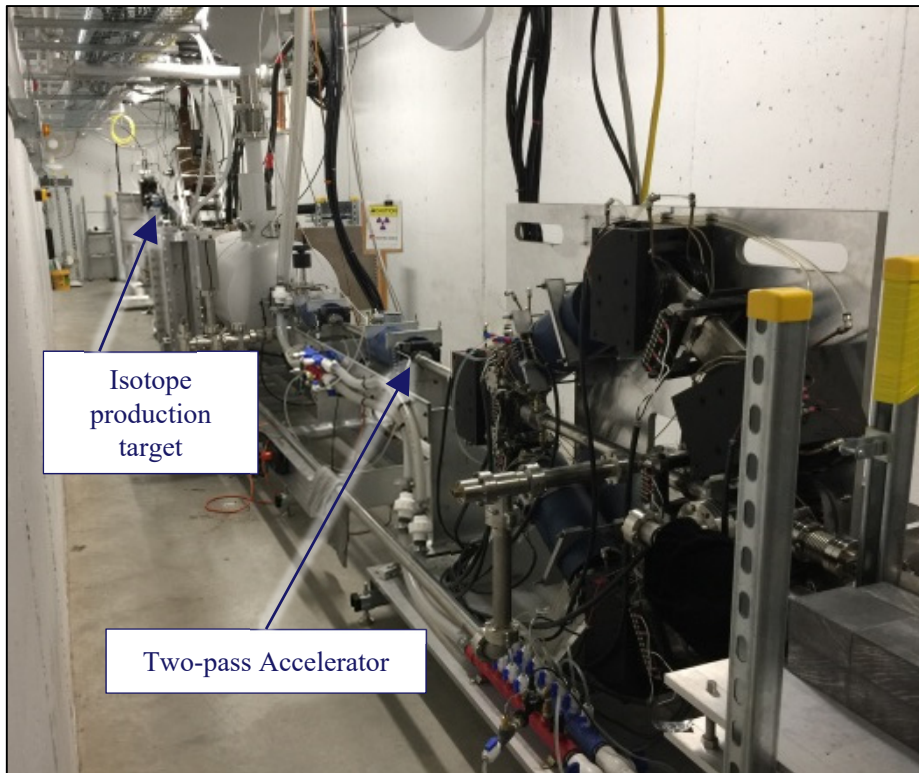
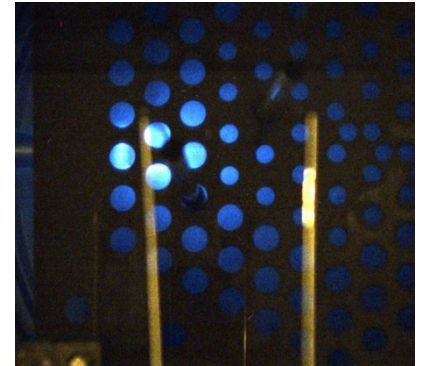
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Isotope Production

Successfully operational fully coupled system:

- Superconducting linac coupled to UTA
- Neutron production verified and validated
- α and β emitters produced





Conclusions



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- As the cost of superconducting RF accelerators continue to come down, commercial applications are increasingly attractive.
- Niowave, Inc. is focused on using SRF electron machines to drive radioisotope production targets and make beta and alpha emitters for medicine.
- WE ARE HIRING FOR A VARIETY OF TECHNICAL AND MANAGEMENT POSITIONS.